Type MS · Compare blood entering and leaving the kidney

Answer 1 of 5 · Question 8 part a.ii · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

8 (a)(i)	• high (blood) pressure (in the glomerulus); • small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule);	Do not allow protein	1 1
(ii)	Any four from: • lower concentration of glucose/oxygen in blood in renal vein; • used in respiration; • less/no urea in renal vein; • urea excreted in urine; • more carbon dioxide in renal vein; • carbon dioxide produced in respiration;	Allow reverse argument throughout	Max 4
(b) (i)	quantity of salt (in food);		1
(ii)	Any two from: • different foods used (which may affect the amount of urea produced/amount of water reabsorbed); • different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance); • the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced);		Max 2
(iii)	use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content);	Allow valid alternatives	1

Total question 8 = 10 marks

Type MS · Compare blood entering and leaving the kidney

Answer 1 of 5 · Question 8 part a.ii · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare blood entering and leaving the kidney

Answer 2 of 5 · Question 8 part a.iii · 2 marks

2020 JAN · Paper 01 · find part a.iii in this mark scheme extract

Number				
8	(a) (i)	X = urethra (1) Y = ureter (1)		2
	(ii)	Any two from: • transports oxygenated blood from the heart (1) • to the body/named organ (1) • maintains high blood pressure (1)		2
	(iii)	Any two from • blood in A/renal artery contains more oxygen/oxygenated (1) • blood in A/renal artery contains more urea (1) • blood in A/renal artery contains more ions (1) • similar concentration of glucose/B renal vein has less glucose (1) • blood in A/renal artery contains less carbon dioxide (1)	Allow reverse argument for B/renal vein	2
	(b) (i)	Any four from: • dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1) • urea/waste/ions diffuse from blood (1) • through dialysis membrane (1) • from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1) • dialysis membrane is partially-permeable(1)	ORA allow selectively permeable membrane	4
	(ii)	• to prevent blood from clotting whilst in the machine (1) • (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1) • (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1)		3

Total for question 8 = 13 marks

Type MS · Compare blood entering and leaving the kidney

Answer 3 of 5 · Question 6 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Question number				
6	(a) (i)	• separating different sized molecules out of the blood/only allows small molecules to pass; • under pressure;		2
	(ii)	• A has more glucose; • A has more urea; • A more amino acids;	ORA vessel B allow references to more oxygen in A and less carbon dioxide in A	3
	(iii)	• Vessel A bring blood to Bowman's capsule/glomerulus; • vessel B has narrow(er) lumen/diameter; • causes increase/higher blood pressure; • to force out molecules;		4
	(b)	• glucose would pass from blood; • respiration affected/reduced; • lack of energy/ATP;		3

Total 12 marks

Type MS · Compare blood entering and leaving the kidney

Answer 3 of 5 · Question 6 part a.ii · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Compare blood entering and leaving the kidney

Answer 4 of 5 · Question 7 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

7 (a) (i)	C; (ureter) *A is a blood vessel* *B is a blood vessel* *D is not attached to kidney*		1
(ii)	A; (bladder) *B is part of the reproductive system* *C is part of the reproductive system* *D leads from the bladder*		1
(b)	• more oxygen; • less carbon dioxide; • more urea;	accept reverse for blood leaving kidney	3
(c) (i)	16.8 - 3.6 = 13.2; 13.2 ÷ 3.6 =3.66; 3.66 × 100 = 366.7%;		3
(ii)	• urea; • large/significant increase in protein-rich diet;		2
(iii)	Any four from • urea, uric acid and ammonia masses are different between the two diets; • ref. creatinine; • amount same/nearly same; • in both diets; • possible result of muscle breakdown;		4

Total 14 marks

Type MS · Compare blood entering and leaving the kidney

Answer 4 of 5 · Question 7 part b · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Compare blood entering and leaving the kidney

Answer 5 of 5 · Question 3 part a.i · 1 mark

2021 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
3	(a) (i)	water;		1
	(ii)	• allows urea to pass; • prevents other larger molecules from passing;	Allow small molecules go through	2
	(iii)	• breaks down urea; • to produce ammonium ions; • to react with/stimulate sensor;	Allow ammonia Allow idea of being picked up/detected by sensor	3
	(b)	• enzyme activity increases/enzyme works faster/higher rate of reaction; • at higher temperatures; • more ammonium ions produced; • higher reading;	Ignore details related to enzyme denaturing / collision theory/kinetic energy ORA for lower temperature	4
	(c)	any two from • can be reused; • products not contaminated; • enzymes more stable; • cost effective justified;	Allow can withstand temperature/pH changes	2

Total 12 marks

Type MS · Compare blood entering and leaving the kidney

Answer 5 of 5 · Question 3 part a.i · 1 mark

2021 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks